

## SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

### ASSESSMENT TOOL 2 – Scenario-Based

|                         |                                                                                                                   |                      |                                              |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------|
| <b>Course Code:</b>     | ITA0532                                                                                                           | <b>Course Title:</b> | Computer Vision                              |
| <b>CO Assessed:</b>     | CO2 – Imitation (BL3)                                                                                             | <b>Assessment:</b>   | Assessment Tool 1 –<br><b>Scenario-Based</b> |
| <b>Weightage:</b>       | 10% of CIA                                                                                                        | <b>Total Marks:</b>  | 100                                          |
| <b>CO2 Statement:</b>   | <i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i> |                      |                                              |
| <b>Student Name:</b>    | Malli Chandana                                                                                                    | <b>Reg. No.:</b>     | 192472250                                    |
| <b>Section Batch:</b> / | BE-CSE-AI/2024-2028                                                                                               | <b>Date:</b>         | 05-08-26                                     |

#### Code of Conduct

*I Malli Chandana(192472250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Malli chandana

## RUBRICS FOR CALCULATION: ASSESSMENT TOOL 2: Scenario-Based

| Criteria                            | Weightage  | Level 4 – Exemplary                                                       | Level 3 – Proficient                               | Level 2 – Developing                              | Level 1 – Beginning                               | Marks |
|-------------------------------------|------------|---------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|-------|
| <b>Understanding of Scenario</b>    | <b>20%</b> | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps           | Partial understanding ; misses key aspects        | Misinterprets or fails to understand the scenario |       |
| <b>Identification of Key Issues</b> | <b>20%</b> | Accurately identifies all relevant issues and factors involved            | Identifies most key issues with minor omissions    | Identifies limited issues; some irrelevant points | Unable to identify key issues                     |       |
| <b>Application of Concepts</b>      | <b>25%</b> | Applies appropriate concepts effectively to address the scenario          | Applies concepts with minor errors                 | Limited or partially correct application          | Unable to apply relevant concepts                 |       |
| <b>Decision Making</b>              | <b>20%</b> | Makes well-reasoned, logical decisions with strong rationale              | Makes reasonable decisions with some justification | Makes weak decisions with limited justification   | No clear decisions made or rationale provided     |       |
| <b>Clarity of Response</b>          | <b>15%</b> | Response is clear, structured, and logically presented                    | Generally clear with minor issues                  | Somewhat unclear or poorly structured             | Disorganized and difficult to understand          |       |

### **30. Industrial Inspection Enhancement**

**An industrial system requires detecting defects in noisy images.**

- (a) Interpret the scenario and define requirements.**
- (b) Identify key issues in detection.**
- (c) Apply an appropriate enhancement and segmentation pipeline.**
- (d) Justify your sequence logically.**
- (e) Present your explanation clearly.**

#### **(a) Interpret the Scenario and Define Requirements**

Industrial inspection is an important application of **Digital Image Processing (DIP)** used in manufacturing industries to automatically identify defective products before they reach customers. Cameras capture images of products moving on a conveyor belt, and image processing techniques analyze these images to detect defects such as scratches, cracks, dents, holes, stains, missing components, and surface irregularities.

However, due to poor lighting, sensor noise, dust, vibrations, and reflections, the captured images often contain distortions that reduce detection accuracy. Therefore, an effective image enhancement and segmentation pipeline is required.

#### **System Requirements**

- Capture high-quality images using industrial cameras.
- Remove different types of noise.
- Improve image brightness and contrast.
- Enhance edges of defective regions.
- Segment defects accurately from the background.
- Detect and classify defects automatically.
- Provide real-time inspection with minimal processing delay.
- Reduce false positives and false negatives.
- Improve product quality and manufacturing efficiency.

#### **(b) Identify Key Issues in Detection**

Several challenges affect defect detection:

##### **1. Noise**

Random noise generated by sensors or the environment can hide small defects.

## 2. Uneven Illumination

Different lighting conditions make segmentation difficult.

## 3. Low Contrast

Defects sometimes have almost the same intensity as the background.

## 4. Reflection and Shadows

Metallic surfaces create reflections that resemble defects.

## 5. Small Defect Size

Tiny cracks and scratches are difficult to identify.

## 6. Motion Blur

Fast-moving conveyor belts may produce blurred images.

## 7. Complex Background

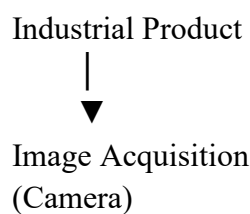
Patterns or textures may be mistaken for defects.

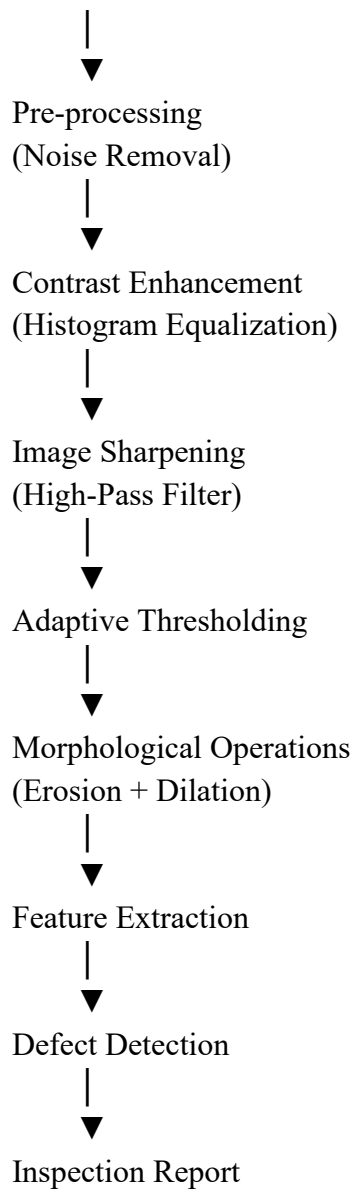
**Table: Problems and Solutions**

| Problem             | Solution                 |
|---------------------|--------------------------|
| Gaussian Noise      | Gaussian Filter          |
| Salt & Pepper Noise | Median Filter            |
| Low Contrast        | Histogram Equalization   |
| Blurred Edges       | High-Pass Filter         |
| Uneven Lighting     | Adaptive Thresholding    |
| Small Noise Regions | Morphological Operations |

## (c) Apply an Appropriate Enhancement and Segmentation Pipeline

### Complete Processing Pipeline





## **Explanation of Each Stage**

### **Step 1 – Image Acquisition**

Images are captured using CCD or CMOS industrial cameras under controlled lighting conditions.

### **Step 2 – Noise Removal**

Median or Gaussian filtering removes unwanted noise while preserving important image features.

### **Step 3 – Contrast Enhancement**

Histogram Equalization stretches the intensity distribution, making hidden defects more visible.

#### **Step 4 – High-Pass Filtering**

Edges and fine details are enhanced to improve defect visibility.

#### **Step 5 – Adaptive Thresholding**

Each local region receives its own threshold, allowing accurate segmentation even under varying illumination.

#### **Step 6 – Morphological Operations**

- **Erosion:** Removes isolated noise.
- **Dilation:** Restores object boundaries.
- **Opening:** Removes small objects.
- **Closing:** Fills small holes.

#### **Step 7 – Feature Extraction**

Important properties such as area, perimeter, shape, texture, and orientation are measured.

#### **Step 8 – Defect Detection**

The extracted features are compared with predefined quality standards to identify defective products.

### **(d) Justify the Sequence Logically**

The order of operations is important because each step improves the effectiveness of the next.

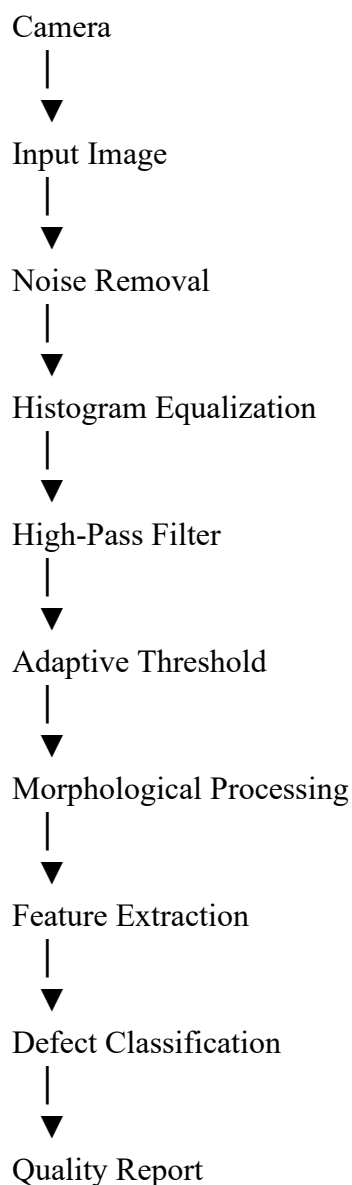
1. **Noise Removal** prevents false edges during sharpening.
2. **Contrast Enhancement** increases the visibility of defects.
3. **High-Pass Filtering** sharpens boundaries.
4. **Adaptive Thresholding** accurately separates foreground from background.
5. **Morphological Processing** removes unwanted pixels and smooths object boundaries.
6. **Feature Extraction** calculates measurements needed for decision-making.
7. **Defect Detection** classifies products as acceptable or defective.

This pipeline produces accurate and reliable inspection results while minimizing processing errors.

### (e) Present the Explanation Clearly

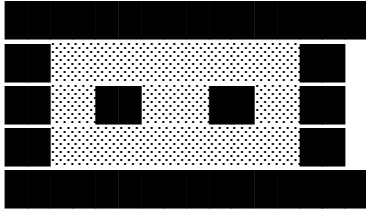
The proposed enhancement and segmentation pipeline provides an efficient solution for industrial quality inspection. By combining noise reduction, contrast enhancement, edge sharpening, adaptive thresholding, and morphological processing, the system accurately detects even small defects. The automated inspection process reduces human error, improves production speed, and ensures consistent product quality. Such systems are widely used in modern smart factories and Industry 4.0 environments.

#### Workflow Diagram

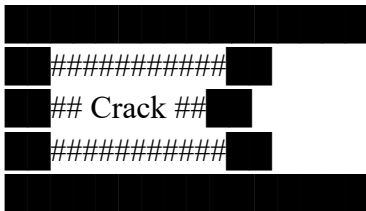


## Example (Before and After Processing)

Original Image



After Enhancement



## Advantages

- High detection accuracy.
- Reduces manual inspection effort.
- Fast real-time processing.
- Detects very small defects.
- Improves manufacturing quality.
- Minimizes production losses.
- Supports automation in Industry 4.0.



## Applications



- **PCB Manufacturing:** Detects missing tracks, solder defects, and broken connections.
- **Textile Industry:** Identifies holes, stains, and weaving defects.
- **Automobile Industry:** Detects scratches, dents, and paint defects.
- **Metal Manufacturing:** Finds cracks, corrosion, and surface imperfections.
- **Food & Pharmaceutical Industry:** Inspects packaging defects and missing labels.
- **Glass Manufacturing:** Detects bubbles, chips, and cracks.

## Conclusion

The proposed enhancement pipeline combines image preprocessing, filtering, segmentation, and feature extraction to achieve reliable defect detection in industrial environments. It improves image quality, enhances defect visibility, and enables accurate automated inspection. Such systems increase production efficiency, reduce operational costs, and ensure that only high-quality products reach customers, making them an essential component of modern automated manufacturing systems.